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Time to Practice: Cluster Quality
Graded Assignment • 15 min

Due Oct 21, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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1. What does a high homogeneity score indicate in cluster analysis? 1 / 1 point

- There is a high degree of overlap between clusters
- The clusters are spread out evenly across the dataset
- Each cluster contains members from multiple classes
- Each cluster contains predominantly only members of a single class

Nice work

Homogeneity measures whether each cluster contains only members of a single class. A high homogeneity score indicates that the clusters are pure, meaning each cluster predominantly consists of a single class of instances.

2. What statistic is commonly used to estimate cluster quality in non-hierarchical clustering methods? 1 / 1 point

- Total number of clusters divided by total data points
- Ratio of total within-groups variance to between-groups variance
- Sum of squared distances from cluster centroids
- Average linkage distance between clusters

Nice work

This ratio is a key statistic used to measure cluster quality, focusing on creating tight clusters that are also well-separated from each other. A higher ratio indicates better clustering by showing that clusters are compact and distinct.

3. If a data scientist does a clustering analysis and observes that the silhouette scores for several items are negative, what does this indicate about the clustering? 1 / 1 point

- Some clusters are too small
- Some items are likely misclustered
- Some clusters are too spread out

Nice work

Negative silhouette scores indicate that the dissimilarity of an item to its own cluster members is greater than to the members of the nearest other cluster, suggesting that the item is in the wrong cluster.

4. Which of the following quality assessment methods is *incorrect* or does *not* capture an important aspect of cluster quality? 1 / 1 point

- Comparing the stability of clustering solutions across different distance measures and/or clustering methods
- Calculating the ratio between the number of clusters and the number of data points
- If we have ground truth data, calculating cluster homogeneity and completeness
- If we don't have ground truth data, computing silhouette scores

Nice work

This statistic may be correlated with cluster quality in some specific cases, but in general, just knowing the average cluster size is not really adequate for capturing key aspects of cluster quality, such as stability, within-cluster compactness, and/or between-cluster separation.

5. A data scientist is trying different numbers of clusters with the k-means algorithm and notices that as they decrease the value of k (the number of clusters) their chosen cluster quality metrics are changing: the between-group variance is decreasing while the within-group variance is increasing. What would be a reasonable action for the data scientist to take next? 1 / 1 point

- Increase the number of clusters
- Keep the number of the clusters the same
- Decrease the number of clusters
- Apply a linear regression model

Nice work

As the number of clusters increases, the centroids are more likely to be closer to the members of their respective clusters, which can decrease the within group variance and increase the between group variance, potentially improving the overall cluster quality.

6. What would be a valid justification for choosing to evaluate the effectiveness of a clustering approach by using a task-based evaluation instead of cluster quality measures like homogeneity and completeness? 1 / 1 point

- A task-based evaluation is typically faster and more efficient than computing metrics like homogeneity, especially with high-dimensional data at scale.
- Ground truth labels are not available for clusters or categories in the data, but an objective quality measure is available for a downstream task that uses the clustering result, e.g. classification.
- Task-based evaluation does not require an objective basis for making comparisons between results.

Nice work

This is correct.